MACH1 Battery Section — 3-Slide Content Pack

Companion to `MACH1_BATTERY_EXECUTIVE_REPORT.md`. Written to be spoken aloud to judges. Trim, reorder, and cut as needed — all figures trace to the executive report and to `/tmp/opencode/mach1/mach1_model.py`.

SLIDE 1 — The Choice and Why It Holds

Headline: LFP + detachable architecture — the only option that survives 2034 homologation, Indian supply, and 8-year duty.

① Why not sodium

- No 45 kWh-class sedan supplier exists
- No cycle data at Indian ambient conditions
- Homologation standards reference **lithium**, not sodium

→ Sodium becomes the **2036–38 second act**, not the primary plan.

② Why LFP

- Domestic supply chain exists — Exide 6 GWh (2026), Agratas (2027), Ola 1.4 GWh
- Best-in-class cycle life
- **Lowest cost per usable kWh**
- No nickel or cobalt price exposure
- **Low charge ceiling is a durability advantage.** Published NMC testing shows degradation sensitivity to how much time a cell spends at high voltage — and that sensitivity **disappears entirely below a 4.06 V ceiling.** LFP runs at ~3.6 V. We are not trying to engineer this property in; our chemistry already sits below the threshold where the problem appears.

③ Why detachable, not structural

- Chemistry swap becomes a **software change, not a new car**
- **AIS-038 Clause 2.48**: a chemistry change needs full REESS retest either way
- But detachable makes the *cell* swappable. Structural CTC locks the chemistry forever
- Cost of that optionality: **₹2–3 Cr / 4–6 months** vs **₹15–25 Cr / 18–24 months**

④ Why the market needs exactly this

- Fleet operators already **multi-source** — Everest runs three Tata models simultaneously

- No platform lock-in anywhere in the segment
- Whoever makes the **best TCO cell** wins the next tender, not whoever owns the platform

⑤ Where the market is going

- **PM E-DRIVE excludes 4W passenger cars** (gazette notification) — the 2025–26 dip is policy, not collapse
- Delhi gazette sets a **staged new-fleet EV ladder: 5% → 50% → 100% over five years**, with full fleet electrification required by **1 April 2030** — and it carries real penalties
- Our 2034 launch lands in the **growth phase**, not the dead zone

> ***Delivery line:*** **"Sodium sounded better on paper. But you cannot homologate a chemistry that has no supplier and no test data. We chose the chemistry we can actually certify — and kept the door open for sodium later by making the pack swappable."**

SLIDE 2 — The Math That Proves It Works

Headline: We didn't estimate. We modelled it against the best published LFP data, and here's what it says.

① Duty cycle — verified, not assumed

- 12.84 kWh/100 km base consumption
- 200 km/day
- **25.68 kWh/day** actual
- ~200 equivalent full charges per year

Our deck originally said 38.28 kWh/day. Real-world data says 25.68 — a 33% correction we made ourselves.

② Pack sizing

43–44 kWh at 48 °C cell. 41 kWh with a proven chiller loop.

③ The finding we lead with

Two peer-reviewed studies. Same LFP chemistry. Same 40 °C. Opposite answers.

Study	Method	Years to 80% SoH
Naumann 2018	Fitted square-root law	13.3 yr
Lewerenz 2017	**Measured linear fade**	**3.7 yr**

Our model **reproduces Lewerenz's own published "3–4 years" exactly.** We are not asserting it — we are checking it.
What we did about it: we didn't pick the flattering number. We reported both, carried the full 0.45–1.00 exponent range, and turned the disagreement into a **6-cell test programme** that resolves it on our chemistry.

④ The exposure nobody models

Resistance ages **4.2× faster** than capacity. At 48 °C: **+101% resistance by year 8 → ~50% of rated power.** A capacity warranty cannot see this. We added a separate warranty clause.

⑤ TCO — already winning, on primary data

City	Ours (EV)	CNG sedan
Delhi	**₹2.55/km**	₹3.15/km
Bengaluru	**₹2.80/km**	₹3.60/km
Mumbai	**₹2.95/km**	₹3.80/km
Hyderabad	**₹2.65/km**	₹3.35/km

Built from OEM fleet price lists, state electricity tariffs, and CNG prices — no contested market-share statistic required.

> ***Delivery line:*** *"Every number on this slide comes from published cell data or state tariffs. Where the literature disagreed, we showed you the disagreement instead of picking the convenient side. And where our own deck was wrong, we corrected it."*

SLIDE 3 — Where the Evidence Comes From

Headline: Peer-reviewed, cell-level, and we know exactly where it does *not* transfer.

① The evidence base

Source	What it gave us
Naumann 2018 — *J. Energy Storage*	LFP calendar model: activation energy, SOC factor
Lewerenz 2017 — *J. Power Sources*	50-cell LFP dataset, linear fade at 40 °C
Simolka 2020 — *J. Applied Electrochemistry*	LFP DoD + temperature post-mortem
Schreiber 2025 — *J. Energy Storage*	Real VW ID.3 duty cycling
Gauthier 2022 — *J. Electrochem. Soc.*	DoD × C-rate × calendar matrix
Eldesoky 2022 — *J. Electrochem. Soc.*	12,000-hr DoD × C-rate × **voltage** × temperature study
Graner 2026 — arXiv	Lab-to-field uncertainty, probability bands
Waldmann 2014 — *J. Power Sources*	Temperature mechanism crossover
Pozzato 2023 — *Joule*	Real-world field behaviour

② What we corrected in our own proposal

- Deck consumption 38.28 → **25.68 kWh/day** (real-world data)
- Charge-rate row **removed** — not a real constraint
- 45 °C ageing claim **restated** on Arrhenius basis
- Reliance/Faradion milestone **replaced**
- Two market statistics **removed** — no primary source exists

③ What we know we don't know

Judges respect this more than false certainty.

- **LFP calendar exponent: 0.45 or 1.0?** Published studies genuinely disagree. This is our single largest open question and it dominates pack size.
- DoD effect: now bounded — best available evidence says the effect is **small** (10–15 mAh across 25→100% DoD over 12,000 hours). What matters is **time at high voltage**, not depth.
- **No LFP-specific cycling dataset exists at our 40–48 °C with partial DoD.** Our numbers extrapolate across chemistries, and we label them as such.
- Louli & Dahn / Harlow 2019: **NMC532, not LFP** — and cycled to a *higher* 4.3 V ceiling than ours. Its 1.6-million-km headline is an upper bound for the field's best benchmark cell, not a number we can claim. Still no LFP DoD matrix in hand.

④ Why that honesty is a strength

The one question that actually drives our answer — the calendar exponent — has two published LFP answers that differ by 3.6×. We show you both, carry the full range, and specify the six-cell test that closes it. Every other assumption is either measured on our chemistry or labelled as borrowed.

> ***Delivery line:*** *"We built this from published cell data, not vendor claims. We found real disagreements in the literature and we showed them to you. And we can tell you exactly which test closes each gap."*

If you're short on time

Must keep

- Slide 2 block ③ — the literature disagreement. This is the credibility anchor.
- Slide 1 block ⑤ — market direction without a contested statistic.
- Slide 3 block ③ — what we don't know.

Can compress

- Slide 1 blocks ② and ③ into one combined block
- Slide 3 block ① into a footnote strip

Never cut

- Slide 2 block ⑤ — the resistance finding. Most original thing in the deck; most reviewers will never have considered it.

Speaker notes — likely judge questions

"Isn't 43 kWh heavy?" At 44 kWh and our verified 25.68 kWh/day, range is ~340 km real-world. Weight sits within the segment for a 45 kWh semi-structural pack, and the 10% EOL reserve means usable capacity drops to ~39 kWh, not 44.

"Why not just buy an LFP pack off the shelf?" Because every published LFP DoD dataset is on a different nickel-based chemistry. We'd be extrapolating across chemistries. The validation programme is what lets us claim the number instead of guessing it.

"Your warranty says 8 years. Your own slide says it might not hold." Yes — deliberately. We show both published answers rather than picking one. That's why the 6-cell test is the first deliverable, and why the warranty is written against a power threshold as well as a capacity threshold.

"How do you know TCO numbers are right?" State electricity tariffs (DERC/BESCOM/TNERC/MERC), CNG prices (IGL/APCPDCL), OEM fleet price lists, and 9.5% fleet finance. All primary. The failure mode we publish with it: below ~150 km/day, or without depot charging, CNG wins.

"Sodium is the future." Agreed — that's why our pack is swappable. We'd change the chemistry without changing the car. But we can't certify it in 2034 with no supplier and no test data.